

Family Resilience Technologies: Designing Collaborative Technologies for Caregiving Coordination in the Children's Hospital

Sarah Nikkhah

School of Informatics and Computing, Indiana University, Indianapolis, IN, USA
snikkhah@iu.edu

ABSTRACT

When a child is hospitalized with a serious illness like cancer, their families must process emotional stress, complicated medical information, and take on new caregiving tasks. Additionally, they must collaborate with each other while maintaining their existing employment and household chores. Computer-Supported Collaborative Work (CSCW) technologies have the ability to enhance family care coordination. However, the challenges to family caregiving coordination in the hospital, particularly during prolonged hospitalizations, are not well studied. In this paper, I describe my proposal and research plan to study the role that collaborative technology can play in augmenting family care collaboration and family resilience during the extended hospitalization period.

CCS CONCEPTS

• **Human-centered computing**; • **Collaborative and social computing**; • **Empirical studies in collaborative and social computing**;

KEYWORDS

Care coordination, Families, Hospitalization, Design Sessions

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1 INTRODUCTION

Each year, the parents of approximately 15,300 kids will hear the words "your child has cancer" [18]. Families with hospitalized children must process a lot of stress and play a vital role in their child's care [6, 15]. Hospitalized children need care and assistance with processing medical information and going through their treatment [7]. Therefore, their families must take new responsibilities such as providing care, processing medical information, getting ready for the extensive and sometimes painful treatments, and face the

fear of losing their child. They must also adjust their daily duties, chores, and jobs to provide care to their hospitalized child.

Previous research on families with hospitalized children shows a lower level of stress and a higher level of communication among family members are significant predictors of long-term health outcomes after hospitalization [5, 13]. Social work and family therapy studies researched family resilience as these families' ability to process and handle stress as a system [14, 19, 20]. However, few technologies are designed to increase family resilience and support the family's communication and collaboration when a child is hospitalized. In this proposal, I will conduct a study on the role that future collaborative technologies can play in supporting family resilience processes and family coordination for families undergoing the high stress of having a child with cancer during the extended hospitalization stay.

There are many Human-Computer Interaction (HCI), and Computer-Supported Cooperative Work (CSCW) studies on the role of technology in supporting coordination and communication within families and between patients, providers, and caregivers. Most of the HCI and CSCW studies on family collaboration concentrate on collaboration within family members in normal settings [2, 8–10], and they do not target families under stress when practices continuously change, and unexpected events occur. Most HCI studies relevant to connected care concentrate on collaboration within patients [1, 12], or patients with providers [11, 16], or patients with caregivers [3, 4, 7]. However, there is a need to understand how collaborative technologies can help family members of hospitalized children (family caregivers) collaborate and coordinate with EACH OTHER during the stressful extended hospitalization period. Family resilience can be used as a lens to understand families' collaborative processes and guide the design of collaborative technologies to support these families in adapting when they are under stress, and their usual routines as a family are constantly changing due to their child's hospitalization. Therefore, there is an opportunity for HCI and CSCW to study the role that collaborative technology can play in supporting family resilience processes for families facing a crisis, such as having a hospitalized child.

2 AIMS

The overall aim of my research is to understand how technologies can support family resilience processes in families undergoing stress. In this proposal, I aim to address this through a specific sociotechnical context: how can collaborative technologies support family resilience processes in families of children with

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extensive hospitalization? Through qualitative interviews and elicitation activities followed by iterative cycles of design, development, and evaluation, I plan to address the following aims in my study.

2.1 Aim 1: Characterizing families' current coordination practices, technology use, and Family Resilience (FR) processes post-diagnosis and during the child's hospitalization stay.

RQ1: What are the current communication and coordination challenges/ practices between family caregivers? RQ2: What is the current family caregiving system's composition, and what is their current technology use? RQ3: What are the common stressors and tensions in communication and coordination of family caregivers?

RQ4: What family resilience processes help these families to tackle the health crisis of having a child with cancer?

I have examined current communication and collaboration practices between a hospitalized child's parenting dyad in my previous work. It helped me characterize some of the parenting dyad's challenges, stressors, coordination, and collaboration practices and revealed some unknowns and areas for more focus, such as technology use and preferences, the other caregivers involved in the family caregiving circle, such as the grandparents, and the family resilience processes. In the summer, I conducted a literature review on collaborative technologies for families and connected care to address the technology use and preference gap. To understand who is involved in the family caregiving circle, I will conduct a survey to identify the family caregiving circle, their involvement in care, and their technology use and preferences. I will merge the findings with a family resilience-focused interview with parents and other caregivers to identify the whole family's current collaboration practices and resilience processes as a system rather than just the parenting dyad. Using concepts of 'family resilience processes' and 'social support theory,' from the family therapy studies, social work, and HCI literature, I will sensitize [17] the concepts in my analysis to characterize family resilience processes and identify potential areas for the design of FR technologies.

3 AIM 2: FURTHER EXPLORING DESIGN OPPORTUNITIES AND ENVISIONING THE DESIGN IMPLICATIONS TO SUPPORT FR PROCESSES FOR FAMILIES WITH HOSPITALIZED CHILDREN.

RQ1: What are the existing tools and technology that families use to communicate and collaborate in the health context? What is the area for more work?

RQ2: What type of design and technology can help families with a hospitalized child coordinate and collaborate? RQ3: To what extent do users feel the app supports different family resilience processes?

Based on the learnings from aim one, I will rapidly design multiple low-fidelity prototypes with different features identified by the interviews, elicitation activities, and surveys. Examples of such prototypes can be calendar, todo, wellness center, tracker, information hub, etc. Then, I will use the concepts and wireframes

to set up co-design sessions with family members to design the system that fits best for their family. Based on the sessions, I will iteratively design increasingly high-fidelity prototypes to support family collaboration. These sessions and design iterations will help me identify design implications to support FR processes and design an interactive prototype to support family caregiving collaboration.

4 AIM3: DEMONSTRATING AND EMPIRICALLY EVALUATING THE TECHNOLOGIES TO SUPPORT FAMILY RESILIENCE ACTIVITIES IN FAMILIES WITH HOSPITALIZED CHILDREN.

RQ1: To what extent do users feel the app supports different family resilience processes? RQ2: Understand care collaboration practices using the app?

RQ3: What are the interactions and patterns of use when families use the FR technology designed to support the caregiving collaboration?

Using the design implications from the previous phase, I will build an interactive prototype of a technology to support family collaboration and FR practices for families with a hospitalized child and will test that through user testing and conducting a pilot study with these families. Using this approach, I can evaluate the design implications found by my study and understand the influence of such technology on FR practices and collaboration activities and explore impacts, possibilities, and challenges for broad implementation of such technology. An example of such a tool might be a multiplatform information huddle with different information channels for family members to communicate about the hospitalized child and other daily activities. (See Table 1 for an overview of methods and objectives).

5 CONTRIBUTIONS

One of the primary outcomes of this work is designing a Family Resilience technology to improve connected care for cancer patients' caregivers. This work also advances knowledge in HCI and CSCW by generating a set of design implications and an example of a designed technology for within family collaboration. In this work, I will study the whole family as a system (rather than just the patient or just the parents) and will examine areas for family-centered health informatics and family collaboration technologies. Unlike most studies on collaborative technologies, in this work, I concentrate on collaboration technologies that support practices within the small and close-knit groups outside the work setting (family members) and not in normal settings (in a crisis, when family members have to face the stress of having a hospitalized child with cancer). Also, this study will provide thick descriptions of family resilience practices across the child's hospitalization journey, such as technology use, communication, and coordination preferences, challenges and common stress factors, coping strategies, and barriers among families with hospitalized children to guide designers of future technologies.

Table 1: Research Methods and Objectives

Specific Aim	Research Questions	Methods	Analysis	Expected Outcomes
Aim1: Characterizing families' current coordination practices, use of technology, and Family Resilience (FR) processes post-diagnosis and during the child's hospitalization stay.	RQ1: What are the current communication and coordination challenges/ practices between family caregivers?	Semi-Structured Interviews with Family Caregivers	Iterative Inductive Coding Thematic Analysis	Characterize current coordination practices
	RQ2: What is the current family caregiving system's composition, and what is their current technology use?	Surveys from Family Caregivers	Regression Analysis Looking for correlations and causations	Amend and validate the findings from interviews, identify who are members of the family caregiving team and their technology use and preferences
	RQ3: What are the common stressors and tensions in communication and coordination of family caregivers?	Elicitation activities: Card Sorting (Stressors, Challenges)	Thematic Analysis	Identify common stressors and their impact on caregiving communication and current FR Processes to tackle stressors
	RQ4: What family resilience processes help these families to tackle the health crisis of having a child with cancer?	ecomaps or care maps FR Interviews with Caregivers and Patients	Deductive Coding	Identify areas for designing FR technologies
Optional additional study to enrich aim one	RQ1: What are the patterns and categories in the text-based communication between parents of hospitalized children?	FR Surveys with Caregivers and Patients	Regression Analysis	Identifying common FR Processes and validating data from the FR interviews.
		Archived Text Transcripts between Caregivers	Contextual Coding Linguistic Analysis Sentiment Analysis	Identifying common trends across time by
	RQ2: Is the text-based communication style relevant to the FR level in families?	Follow up phone call	Thematic analysis	characterize text-based Coordination practices across time
		Comparative Study	Comparative analysis between FR survey results and text-based analysis	Identify indications of tensions and stress or negative sentiments Identify successful practices based on FR level and text-based communication patterns
Aim2: Exploring design opportunities and envisioning the design implications to support FR processes for families with hospitalized children.	RQ1: What are the existing tools and technology that families use to communicate and collaborate in the health context? What is the area for more work?	Design feature exploration	Literature review	Find design opportunities
		Participatory Design Sessions with Caregivers and Patients and generating empathy maps	Thematic analysis	Design implications for FR Technologies Design Implications for supporting caregiving practices and care coordination
	RQ2: What type of design and technology can help families with a hospitalized child coordinate and collaborate?	Generate static lo-fi prototypes Test with Design Experts		Challenges in designing for family caregivers
		Generate static hi-fi prototypes	Heuristic evaluations	Guidelines in conducting human-centered research with family caregiver's population
		Test with Caregivers and Patients	Concurrent think-aloud Retrospective think-aloud	
		Interactive Prototyping		
		Test with Caregivers and Patients		

Table 1: (Continued)

Aim3: Demonstrating and empirically evaluating the technologies to support FR activities in families with a hospitalized child.	RQ1: To what extent do users feel the app supports different family resilience processes?	Design a pilot study	Contextual Coding on the texts communicated in the app and transcriptions of video or audio messages or calls in the app	An instance of Functional technology to Support FR Processes and collaborative caregiving
	RQ2: Understand care collaboration practices using the app?	Pre-interview to introduce them to the app		Rich descriptions on the FR technology use
		Ask them to use the app		
		Interview them after they use the app/ follow up call	Thematic coding on the feedback they provided in the follow-up call	Learnings and feedback from family caregivers
	RQ3: What are the interactions and patterns of use when families use the FR technology designed to support the caregiving collaboration?	Asses perceived influence on FR using a measurement survey	Regression analysis of the survey	Design guidelines to support FR technologies and family caregiving collaboration technologies
		Participants in all activities will be Caregivers and Patients	Look for relationships between the use of the app and the FR processes from the participants' perspective.	Find family caregiver's perspective of to what extent the app influences on FR processes
				Find the potentials for FR technologies

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